Project and Hackathon Assignment

Project Baseline

By now, you should have your `k5_xbox` platform setup functional in both **simulation** and **FPGA** environments. You have also been provided with the following reference implementations for a hardware-accelerated Sudoku solver.

sudx_basic

`sudx_basic` is an application that provides hardware acceleration for selected functionality of the software Sudoku solver.

sudx_scan

`sudx_scan` is an integrated, fully hardware-accelerated, scan-based Sudoku-solving algorithm running on the `k5_xbox` platform.

The solving algorithm is similar to `sudx_basic`, but the actual solving is performed entirely in hardware. The software serves only as the user interface and as a wrapper for invoking the hardware accelerator.

While `sudx_scan` performs significantly better than `sudx_basic`, it still require a large number of cycles to solve complicated boards, on the order of **50 million cycles** for the `hard1` board. On the other hand, it can be synthesized to a relatively high operating frequency of **above 50 MHz**.

claude_mrv

`claude_mrv` is a standalone, fully hardware-based solution and testbench implementing the **MRV (Minimum Remaining Values)** Sudoku-solving algorithm.

This solution requires dramatically fewer cycles, on the order of **1,000 cycles** to solve the `hard1` board. However, it synthesizes to a relatively poor maximum frequency, on the order of **5 MHz**.

Unlike `sudx_scan`, this reference implementation is not integrated into the `k5_xbox` platform. However, you should be able to relatively easily utilize it to create a `sudx_mrv` accelerator and corresponding C driver code, based on the `sudx_scan` reference that you already have.

The Project Challenge

The primary goal of the project is to provide an **enhanced hardware-accelerated Sudoku application** running on the `k5_xbox` platform, in both simulation and FPGA, with **significantly reduced solving time**.

The Sudoku solving time is determined by the reported cycle count and multiplied by the accelerator achieved clock period, which can be expressed as:

> **Solve time (μs) = cycle count / maximum frequency (MHz)**

```
solve_time_us = cycle_count / max_freq_mhz
```

To neutralize the platform infrastructure speed bottleneck, the `max_freq_mhz` value should be taken from the **standalone accelerator synthesis** using the `qsyn_xlr` utility, rather than from the full-design `comp_fpga` result.

Developing Your Acceleration Version

You may select one of the reference implementations above as your baseline and attempt to improve its solving time according to the formula above.

This can be achieved by:

- Reducing the achieved **cycle count**
- Increasing the achieved **maximum frequency**
- Ideally, improving both

You may also choose to develop your own complete acceleration solution, including both **hardware and software components**, as long as it fits within the `k5_xbox` platform infrastructure.

> **AI assistance:** You may use AI tools to assist with your development, as long as you reasonably understand the generated code and can explain your solution.

We recognize that the different baseline choices provide significantly different performance ranges. Therefore, much of the project evaluation will be **relative to the selected baseline and the effort made to improve its performance**.

However, to some limited extent, we will also take into consideration:

- The complexity and sophistication of the provided solution
- The absolute performance achieved

The Hackathon Event Challenge

The last day of the course will be dedicated to the **Hackathon Challenge**.

COOL: I just noticed that this happens to be on **9/9**, which is a great date for a 9×9 Sudoku Solver Hackathon! 😊

Shortly before the Hackathon event, somewhere between a few hours and two days (to be determined) we will reveal a variant of the Sudoku challenge.

You will then be required to adapt your project solution, or another solution of your choice, to solve the provided variant.

If you want to guess what such a variant might look like, you can easily search online for Sudoku variant or so.

The selected variant will not be among the most complex ones you may find, but it will also not be trivial. If you understand your project solution well, adapting it to the provided variant should be realistic within the available Hackathon time.

Hackathon Competition and Recognition

The Hackathon will also include a competition element. This part does not directly impact the course grade and will be further based on the absolute performance achieved.

The top three places will receive awards. (We cannot yet commit to prizes.) Additional recognition may also be awarded for achievements such as:

- Special creativity
- Exceptional dedication
- Other notable contributions or accomplishments

All participants will receive a digital **"Successful Hackathon Participant" certificate**, which can be a nice addition to your LinkedIn or profile and resume.

Overall Achievement Grading

> **Note:** The grading breakdown below is not yet final and may be subject to change.

Category	Item	Weight
Baseline	SUDX baseline functional in simulation	5%
	SUDX baseline functional on FPGA	10%
Project	Solved in simulation	5%
	Demonstrated on FPGA	10%
	Solution efficiency and performance	30%
Hackathon	Variant solved in simulation	10%
	Variant demonstrated on FPGA	10%
	Variant efficiency and performance	5%
Submission	Overall code quality and understanding	10%
	Project & Hackathon report	5%
	Total	**100%**

> **Notice:** The Project and Hackathon grade constitutes **40% of the overall course grade**.

Important: Student Cooperation Guidelines

The Project and Hackathon work is **individual work**, not pair or group work. You are, of course, encouraged to consult with other students and definitely with the course staff to discuss ideas. However, you are expected to deliver a **reasonably unique solution**, and your work will be evaluated individually based on your own implementation and results.

Looking forward to a fruitful project, an exciting Hackathon, and a great course closure!

DDP26-Summer Staff